

Trainings ▾

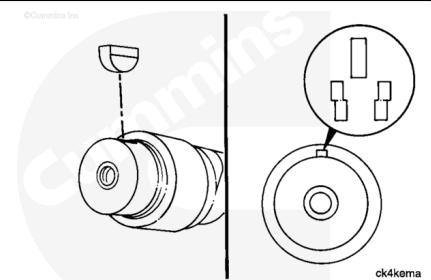
General Information

Camshaft keys are available in different sizes (amount of offset). The injection timing is controlled by:

- The selection of the camshaft key
- The direction of the key offset when compared to the camshaft gear rotation
- The amount of offset

Check and record the direction of the key offset when compared to the camshaft gear rotation (same or opposite).

Note : Engines with electronically actuated injectors do not use off set timing keys.

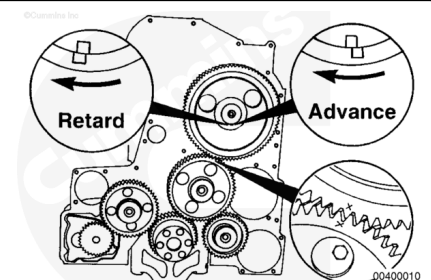


The camshaft key provides a means of indexing the camshaft with the gear.

Offset keys allow the camshaft profile to be rotated slightly while the gear train timing remains the same.

The more the top of the offset is moved in the direction of the camshaft normal rotation, the more the injection timing will be retarded. The push rod travel numerical value will increase.

Note : The direction of normal rotation on a QSK19 engine crankshaft is clockwise as viewed from the front of the engine.



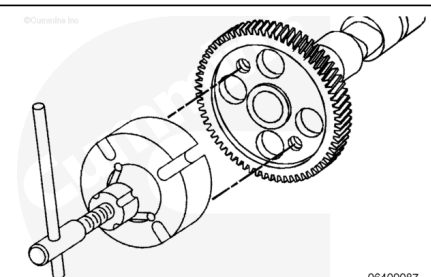
Remove

Use a camshaft gear puller, Part Number 3824900, to remove camshaft gear.

Attach the puller to the gear. Remove the gear.

Remove the key.

Remove the puller.

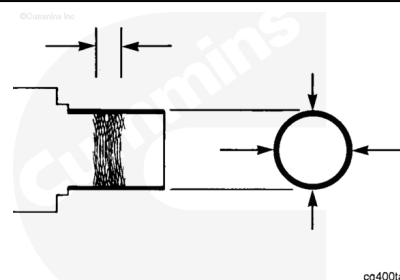


Clean and Inspect for Reuse

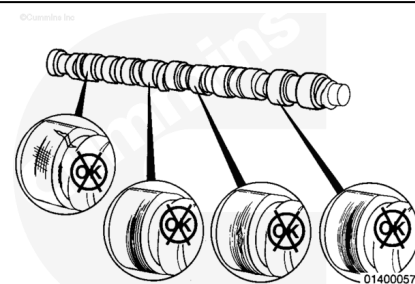
Inspect the camshaft and camshaft gear for reuse. Check for fretting damage. The camshaft **must** be replaced if the fretting damage is more than 3 mm [1/8 inch] wide.

Measure the outside diameter of the camshaft nose.

mm		in
57.200	MIN	2.2520
57.210	MAX	2.2524



Inspect the camshaft. Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/20/20-001-008-tr.html>).

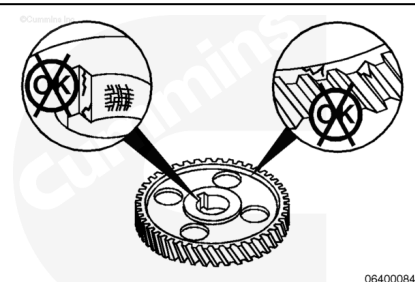


Inspect the camshaft gear for cracks, chipped or broken teeth.

Inspect the bore of the gear for fretting or burrs.

Measure the inside diameter

mm		in
57.125	MIN	2.2490
57.150	MAX	2.2500

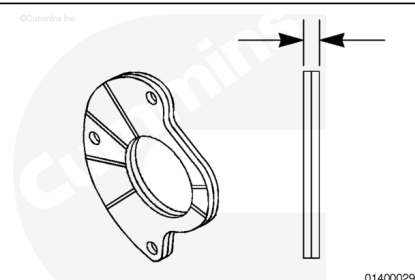


The gear **must** be replaced if the inside diameter is **not** within specification.

Inspect the thrust bearing for damage.

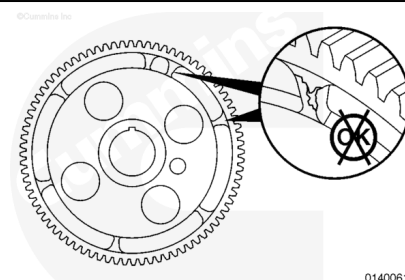
Measure the thickness.

mm		in
9.34	MIN	0.368
9.44	MAX	0.372



with Electronically Actuated Injector

Inspect the back side of the gear for cracks, chipped, or broken bosses that the cam position sensor reads. The gear **must** be replaced if the bosses are damaged.



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Magnetic Crack Inspect

⚠ CAUTION ⚠

Use a copper braid contact that has neoprene covers to reduce the possibility of burning the teeth of the gear.

If the gear contains a keyway, position the gear so that the keyway is pointing toward one of the contacts.

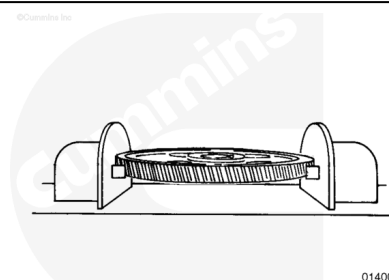
Use a magnetic particle testing machine.

Use the residual method and apply head shot amperage.

Adjust the amperage to the value specified in the table.

Amperage for Gear Sizes	
Gear Outside Diameter	Direct Current Amperage
Less than 101 mm [4 in]	1000
101 to 203 mm [4 to 8 in]	1500
Greater than 203 mm [8 in]	2000

Check the camshaft gear for cracks.

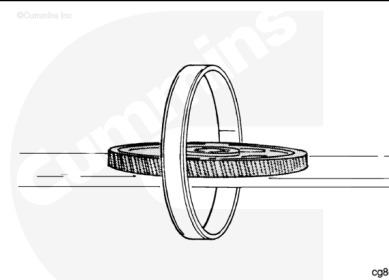


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If the gear contains a keyway, position the gear so that the keyway is pointing toward the coil.

Use the residual method and apply coil shot amperage.

Adjust the amperage to the value specified in the table.



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Amperage for Gear Sizes	
Gear Outside Diameter	Ampere Turns
Less than 101 mm [4 in]	4000
101 to 203 mm [4 to 8 in]	6000
Greater than 203 mm [8 in]	8000

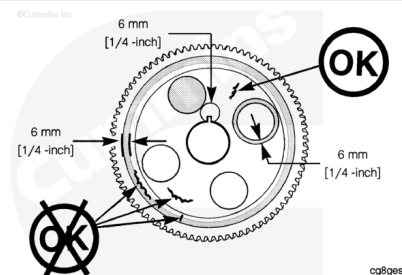
Gear Outside Diameter	Ampere Turns
Less than 101 mm [4 in]	4000
101 to 203 mm [4 to 8 in]	6000
Greater than 203 mm [8 in]	8000

Check the camshaft gear fro cracks.

The gear must be demagnetized completely and cleaned thoroughly. Any small particles will cause engine damage.

Limits of Acceptance - Machined Surfaces.

- There is an open indication.
- There is an indication below the surface that is in the shaded area, shown in the graphic.
- There is an indication below the surface that is longer than 6 mm [0.250 in].



- There is a open indication that is in a circumferential direction.
- There is an open indication that is longer than 9.5 mm [0.375 in].
- There is an indication below the surface that is in the shaded area shown in the graphic.

Demagnetize the gear.

Install

Note : Engines with electronically actuated injectors do not use off set timing keys.

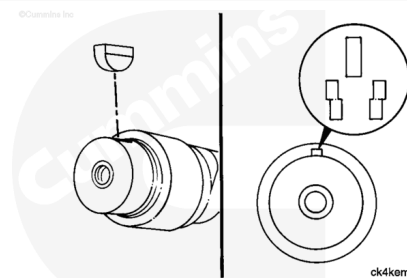
Note : The camshaft gear must be removed to adjust the injection timing. If the camshaft was replaced, do not install the camshaft gear until the injection timing has been verified

Use a slip-fit gear to adjust injection timing. Refer to Procedure 006-025 (/qs3/pubsys2/xml/en/procedures/20/20-006-025-tr.html).

The following table lists the recommended camshaft key for each timing code. Machining tolerances on the camshaft and the gear can cause the injection timing **not** to be within specifications. If the timing is **not** within specifications. See Section 0 for instructions to choose an alternate camshaft key.

The amount of offset is based on 5.1613 mm (0.2032 inch) of piston travel.

Timing Code	Camshaft Key	Amount of Offset
JF	216294	0.279 Inj.
KG	S-302	0.320 Inj.



⚠ WARNING ⚠

To reduce the possibility of severe burns wear protective clothing when heating the gear.

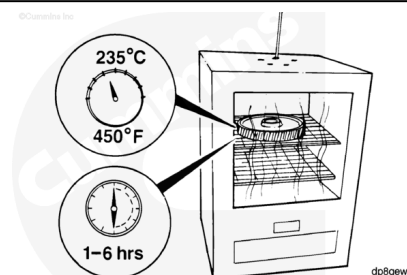
⚠ CAUTION ⚠

Do not exceed the specified time or the temperature. Damage to the gear and the gear teeth will result.

⚠ CAUTION ⚠

Do not attempt to install the gear without using heat.

Use an oven with the temperature adjusted to 235°C [455°F]. Heat the gear in the oven for a **minimum** of one hour but **not** more than a maximum of six hours. The inside diameter of the gear will become larger and will simplify the installation.

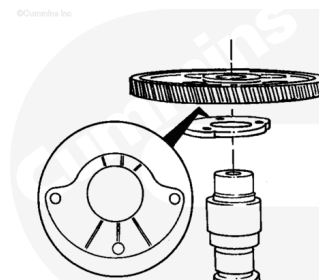


Note : The timing marks **must** be visible on the front of the gear after it is installed. The keyway in the gear **must** be aligned with the key.

Install the thrust plate. Install the camshaft gear.

⚠ CAUTION ⚠

Allow the gear to air cool. Do not use water or oil to reduce the cooling time. It will cause the gear to crack.



Use a feeler gauge. Measure the camshaft thrust clearance.

mm		in
0.15	MIN	0.006
0.33	MAX	0.013

If the clearance is **not** within specification the gear **must** be removed.

